

Program : Diploma in Information Technology	
Course Code : 3263	Course Title: Fundamentals of Web Development
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To develop web pages use Hyper Text Mark-up Language (HTML), Cascading Style Sheet (CSS) and Client-Side scripting Language-JavaScript.

Course Pre-requisites:

Topic	Course Code	Course name	Semester
Knowledge of basics of Computer		Problem Solving & Programming	2
Basic Knowledge of IT skills and basic problem solving		Introduction to IT Systems Lab	2

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concepts Internet, world wide web, client server architecture, and http communication.	10	Understanding
CO2	Develop web pages using HTML.	11	Applying
CO3	Develop web pages using HTML5 and CSS3.	13	Applying
CO4	Illustrate the concepts of JavaScript and validate web pages using JavaScript.	9	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						

CO3	3						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concepts Internet, world wide web, client server architecture, and http communication.		
M1.01	Define Internet and world wide web, web browser and world web addresses (IP, Domain, URL),	2	Remembering
M1.02	Summarize Client server architecture, web browser and web server communication.	2	Understanding
M1.03	Outline Types of Networks Define web protocols- (FTP, DNS, DHCP, HTTP)	1	Remembering
M1.04	Summarize http features, http architecture, http request, response	1	Understanding
M1.05	Define JSON	1	Remembering
M1.06	Summarize JSON with example	1	Understanding
M1.07	Define XML	1	Remembering
M1.07	Summarize XML with example	1	Understanding
Contents: Define Internet - World Wide Web - Web Browser-Web Server and Web Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Server communication - Web protocols. HTTP – Features, Architecture, http Request and Response. Data formats for HTTP Communication –JSON, XML.			
CO2	Develop web pages using HTML.		
M2.01	Define HTML, show needs and history of HTML, Show basic structure of HTML	1	Remembering
M2.02	Summarize head, title and metadata in HTML Summarize heading tags in HTML	1	Understanding
M2.03	Summarize paragraph and line break tags in HTML Summarize Lists in HTML	1	Understanding
M2.04	Summarize Tables in HTML Summarize Image tag and its attributes in HTML Summarize hyperlinks in HTML	2	Understanding
M2.05	Summarize Form tag HTML	1	Understanding

M2.06	Summarize Form elements in HTML (Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons.)	2	Understanding
M2.07	Develop a web page using all the above-mentioned HTML elements.	3	Applying
	Series Test – I	1	

Contents:

HTML: -Describe HTML, Needs and History of HTML, Basic structure of HTML page

Basic Mark-up: -Headings - Paragraphs -line breaks –Lists – Tables-Image Tag and Attributes– Hyperlinks-Form.

Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons.

CO3	Develop web pages using HTML5 and CSS3.		
M3.01	Define HTML 5, Summarize HTML 5 API features	1	Remembering
M3.02	Summarize HTML 5 semantic elements (header, footer, nav, article, div, span, section, figure, figcaption, main, aside)	2	Understanding
M3.03	Summarize HTML 5 Media types (Audio and Video)	1	Understanding
M3.04	Summarize canvas and SVG in HTML 5	1	Understanding
M3.05	Define CSS, List types of CSS	1	Remembering
M3.06	Define CSS3, Summarize CSS3 selectors	1	Remembering
M3.07	Summarize CSS3 borders	1	Understanding
M3.08	Summarize CSS3 colors	1	Understanding
M3.09	Summarize CSS3 backgrounds	1	Understanding
M3.10	Develop a web page using HTML 5 and CSS 3	3	Applying

Contents:

HTML 5: -HTML5 API features-New Semantic Elements in HTML5: -header, footer, nav, article, div, span, section, figure, figcaption, main, aside.

HTML5 Media Types, -Canvas -Scalable Vector graphics.

CSS3: -Define CSS-Types of CSS-Define CSS3-CSS3 Selectors, borders, backgrounds and colors.

CO4	Illustrate the concepts of JavaScript and validate web pages using JavaScript.		
M4.01	Define client-side scripting language, Define JavaScript	1	Remembering
M4.02	Summarize JavaScript syntax, Summarize	1	Remembering

	variables in JavaScript		
M4.03	Summarize decision control statements and loops in JavaScript	3	Understanding
M4.04	Summarize arrays in JavaScript	1	Understanding.
M4.05	Define DOM, Summarize Accessing DOM elements in JavaScript	1	Remembering
M4.06	Summarize form validation using JavaScript	2	Understanding.
	Series Test – II	1	
Contents: JAVASCRIPT: -Define Client-Side Scripting, Define JavaScript. JavaScript syntax- variables in JavaScript - Decision Control Statements in JavaScript – Loops in JavaScript - Arrays in JavaScript. Document Object Model, Accessing DOM elements in JavaScript. Validation of Forms using JavaScript.			

Text / Reference

T/R	Book Title/Author
T1	Julie C Meloni, Sams Teach Yourself HTML, CSS and JavaScript All in One
T2	Lura Iemany, Rafe Colburn, Jennifer Kyrnin-BPB Publications - Mastering HTML, CSS and JavaScript web publishing

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/html/default.asp
2	https://www.w3schools.com/css/default.asp
3	https://www.w3schools.com/js/default.asp
4	https://www.tutorialspoint.com/html5/index.htm
5	https://www.tutorialspoint.com/css/css3_tutorial.htm